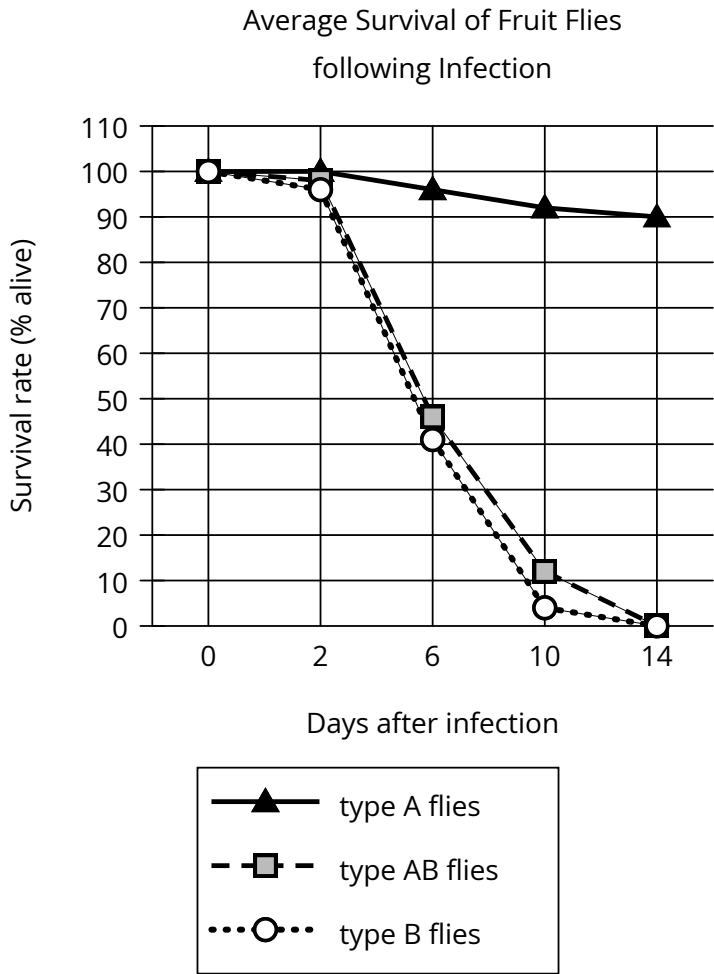


Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In a study of the evolution of *DptA* and *DptB*—*Dipteracin* genes encoding antimicrobial peptides that combat pathogens and foster beneficial microbes in fruit flies (*Drosophila*)—researchers assessed *Drosophila melanogaster* resistance to pathogenic infections by *Providencia rettgeri* and *Acetobacter sicerae*, bacteria common in the flies’ environments. Subjects included flies identified by mutations silencing *DptA*, *DptB*, or both *DptA* and *DptB* (termed types A, B, and AB, respectively). In conjunction with the observation that resistance to *P. rettgeri* correlates with *DptA* activity but is not significantly affected by *DptB* activity, data in the graph of survival rates post-*A. sicerae* infection suggest that _____

Which completion of the text is best supported by data in the graph?

Answer

- A. *DptA* confers defense against *A. sicerae* regardless of the presence of *DptB*.
- B. *DptB* protects against only one bacteria species, whereas *DptA* protects against multiple species.
- C. *DptB* may have developed as a specific defense against *A. sicerae*.
- D. defense against *A. sicerae* is strongest when both *DptA* and *DptB* are present.